

CBSE Practice Worksheet

Class 6 | Science | Difficulty: Hard | Set D

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Which quantum number determines the shape of an orbital?
(A) Principal
(B) Azimuthal
(C) Magnetic
(D) Spin
2. What is the coordination number in FCC structure?
(A) 6
(B) 8
(C) 12
(D) 14
3. What is the hybridization of carbon in methane?
(A) sp
(B) sp^2
(C) sp^3
(D) sp^3d
4. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1
(D) 0 only
5. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s
(D) 25 km/s
6. Which enzyme is responsible for DNA replication?
(A) DNA ligase
(B) DNA polymerase
(C) Helicase
(D) Primase
7. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%
(D) 75%
8. What is the oxidation state of Mn in $KMnO_4$?
(A) +2
(B) +4
(C) +6

(D) +7

9. Which reaction is endothermic?

- (A) Combustion
- (B) Neutralization
- (C) Photosynthesis
- (D) Respiration

10. Which organelle is involved in programmed cell death?

- (A) Lysosome
- (B) Mitochondria
- (C) Peroxisome
- (D) Ribosome

Section B: Fill in the Blanks (1 mark each)

- 11. The uncertainty principle was given by ____.
- 12. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.
- 13. The process of splitting water during photosynthesis is called ____.
- 14. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.
- 15. The Krebs cycle occurs in the ____ of mitochondria.
- 16. In $\text{S}_\text{N}2$ reactions, the rate depends on both ____ and nucleophile.
- 17. For a reversible process, entropy change of the universe is ____.
- 18. The band gap in semiconductors is typically ____ eV.

Section C: Short Answer Questions (2 marks each)

- 19. Explain Le Chatelier's principle with examples.
- 20. Describe the light-dependent and light-independent reactions of photosynthesis.
- 21. Derive the lens formula and explain its applications.
- 22. Explain the structure and working of a nuclear reactor.
- 23. Derive the equation for kinetic energy of gases from kinetic theory.
- 24. Explain hybridization with examples of sp , sp^2 , and sp^3 .
- 25. Derive the expression for electric field due to a dipole.

--- End of Worksheet ---

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